

Problem 1:

Gini:

$$G(p_i) = 1 - \sum_{i=1}^n p_i^2$$

$n=2$ (binary)

$$G(p_1, p_2) = 1 - (p_1^2 + p_2^2)$$

$$= 1 - p_1^2 - p_2^2$$

dG

$$p_1 = \frac{N_1}{N}, \quad p_2 = 1 - \frac{N_1}{N} = 1 - p_1$$

$$G(p) = 1 - p^2 - (1-p)^2$$

$$= 1 - p^2 - (1 - 2p + p^2) = 2p - 2p^2$$

$$= 2p(1-p)$$

$$\Rightarrow 2p(1-p) = 0$$

$$p = 0 \text{ and } p = 1$$

$$\frac{dG}{dp} = 2 - 4p = 0$$

$$p = \frac{1}{2}$$

Shannon entropy:

$$S = - \sum_{i=1}^n p_i \log_2(p_i)$$

$$= -[p_1 \log_2(p_1) + p_2 (\log_2(p_2))]$$

$$= -p_1 \log_2(p_1) - p_2 \log_2(p_2)$$

$$= -p_1 \log p_1 - (1-p_1) \log(1-p_1)$$

$$S(p) = -p \log p - (1-p) \log(1-p)$$

$$= -p \log p - \log(1-p) + p \log(1-p)$$

$$= p \left(\log\left(\frac{1-p}{p}\right) \right) - \log(1-p)$$

$$S(0) = 0 - (1-0) \log(1-0) \quad \left| \quad S(1) = -\log 1 = 0 \right.$$

$$= -\log 1$$

$$= 0$$

$$\frac{dS}{dp} = \cancel{p} \frac{\cancel{p}}{1-p} \left(\frac{-1}{\cancel{p^2}} \right) + \log\left(\frac{1-p}{p}\right) - \frac{1}{1-p} (-1)$$

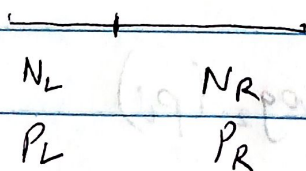
$$(9-9')9 = \frac{-1}{\cancel{1-p}} + \frac{1}{\cancel{1-p}} + \log\left(\frac{1-p}{p}\right) = 0$$

$$2^0 = \frac{1-p}{p}$$

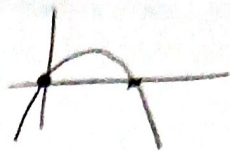
$$p = 1-p$$

$$p = \frac{1}{2}$$

(b)



$$G_{\text{split}} = \frac{N_L}{N} G(p_L) + \frac{N_R}{N} G(p_R)$$



→ Concavity of $G(p)$

→ Show $G_{\text{split}} \leq G(p)$

$$G(p) = 1 - \sum_{i=1}^n p_i^2$$

$$\frac{N_L}{N} p_L + \frac{N_R}{N} p_R = p$$

$$\lambda p_L + (1-\lambda) p_R = p$$

From wolfram alpha: $f\left[\sum_{i=1}^n p_i x_i\right] \geq \sum_{i=1}^n p_i f(x_i)$ if f is concave

$$G\left(\sum_{i=1}^2 p_i x_i\right) \geq \sum_{i=1}^2 p_i G(x_i) \quad \left| \begin{array}{l} x_i \rightarrow \text{classes} \\ p_i \rightarrow \text{weights} \end{array} \right. \quad (?)$$

$$G(p_1 x_1 + p_2 x_2) \geq p_1 G(x_1) + p_2 G(x_2)$$

$$G\left(p_L \frac{N_L}{N} + p_R \frac{N_R}{N}\right) \geq p_L G\left(\frac{N_L}{N}\right) + p_R G\left(\frac{N_R}{N}\right)$$

$$G\left(\frac{N_L}{N} p_L + \frac{N_R}{N} p_R\right) \geq \frac{N_L}{N} G(p_L) + \frac{N_R}{N} G(p_R)$$

$$\frac{N_L}{N} G(p_L) + \frac{N_R}{N} G(p_R) \leq G\left(\frac{N_L}{N} p_L + \frac{N_R}{N} p_R\right)$$

$$\therefore G_{\text{split}} \leq G(p)$$

$$(14) \quad G_{\text{split}} = \frac{N_L}{N} G(p_L) + \frac{N_R}{N} G(p_R)$$

Split A: p_L p_R

class 0, class 1

~~$$G_{\text{split-A}} = \frac{12}{20} G_A\left(\frac{12}{20}\right) + \frac{8}{20} G_A\left(\frac{8}{20}\right)$$~~

$$G_{\text{split-A}} = \frac{12}{20} G_A\left(\frac{2}{12}\right) + \frac{8}{20} G_A\left(\frac{6}{8}\right)$$

$$G_A\left(\frac{2}{12}\right) = 1 - p = 2p(1-p)$$

$$= 2\left(\frac{2}{12}\right)\left(1 - \frac{2}{12}\right)$$

$$= \frac{4}{12} \left(\frac{10}{12} \right) = \frac{40}{144}$$

$$G_A\left(\frac{6}{8}\right) = 2\left(\frac{6}{8}\right)\left(1 - \frac{6}{8}\right) = \frac{12}{8} \left(\frac{8-6}{8} \right) = \frac{12}{8} \left(\frac{2}{8} \right) = \frac{24}{64}$$

$$G_{\text{split-A}} = \frac{12}{20} \left(\frac{40}{144} \right) + \frac{8}{20} \left(\frac{24}{64} \right)$$

$$= 0.167 + 0.15 = 0.317$$

Split A is better

because it has more points in the left child,

Split B is

mixed perfectly.

~~$$G_{\text{split-B}} = \frac{12}{20} G_B\left(\frac{6}{12}\right) + \frac{8}{20} G_B\left(\frac{2}{8}\right)$$~~

$$G_B\left(\frac{6}{12}\right) = 2\left(\frac{1}{2}\right)\left(1 - \frac{1}{2}\right) = \frac{1}{2}$$

$$G_B\left(\frac{1}{4}\right) = 2\left(\frac{1}{4}\right)\left(\frac{3}{4}\right) = \frac{3}{8}$$

$$G_{\text{split-B}} = \frac{12}{20} \left(\frac{1}{2} \right) + \frac{8}{20} \left(\frac{3}{8} \right)$$

$$= \frac{9}{20} = 0.45$$